

Michael Vaden

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, COLLEGE OF COMPUTING

Atlanta, GA

B.S. Computer Science

Expected: May 2026

- **Concentrations:** Intelligence/Systems and Architecture

GPA: 3.54/4.0

- **Coursework:** Applied Combinatorics, Automata and Complexity, Intro Perception and Robotics, Design and Analysis of Algorithms, Systems and Networks

SKILLS

Languages: Rust, Go, C++, C, Python, JavaScript, TypeScript, SQL, Java

Libraries/Frameworks: NumPy, TensorFlow, React.js, Next.js, FastAPI, Django, LLVM, Node.js, Fiber, Selenium, OpenGL, WebGL

Technologies/Tools: Git, UI, Unix, Make/CMake, Blender, Microsoft Power Suite, Adobe Creative Suite, Tableau, Teradata

Cloud/DevOps: AWS (Bedrock, S3, Lambda, Glue, Lake Formation, Athena, Redshift), GCP (Cloud Functions, Firestore), Docker

EXPERIENCE

Delta

Atlanta, GA

Data Co-Op, Tools and Technology

September 2023 – May 2025

- Architected AWS-based data pipelines integrating S3, Athena, Lambda, Glue, and Lake Formation aligned with the division's data lake initiatives enabling automated ingestion, transformation, and querying of multi-terabyte datasets
- Developed standardized ETL procedures with Teradata, SAS, and Athena, increasing data accuracy by 50%
- Designed and deployed a validation and loading pipeline with Lambda and Athena to process vendor-provided denied boarding compensation data, detecting duplicate records and flagging discrepancies in exchange rates before invoice confirmation
- Created and deployed analytical dashboards to deliver real-time business intelligence for cross-departmental stakeholders
- Collaborated with data engineers and business analysts to optimize SQL queries, enforce data governance, and align cloud data architecture with enterprise standards

Georgia Tech IFC

Atlanta, GA

Web Developer

September 2024 – November 2024

- Designed and developed a modern, mobile-responsive website for the Georgia Tech Interfraternity Council with Next.js
- Deployed and maintained the site via custom NGINX configuration on a Digital Ocean Droplet

University of Georgia

Athens, GA

Software Engineering Intern

January 2022 – December 2022

- Implemented a Django-based web application for researchers to manage research groups and provision resources on the university's HPC cluster, Sapelo2
- Integrated CAS authentication for SSO login via the university's security procedures
- Integrated role-based access control to streamline user management across multiple research projects
- Developed automated provisioning workflows with Slurm job scheduler APIs, reducing allocation time and improving ease-of-use in resource provisioning
- Implemented email notification system for notifications and account verification via SMTP
- Collaborated with HPC administrators to optimize cluster onboarding and improve documentation for faculty and graduate researchers

Black Dog Customs

Madison, GA

Electrical/Fabrication Specialist

May 2019 – May 2021

- Designed and manufactured custom automotive electrical systems, including wiring harnesses, switch panels, dashboards, audio systems, and control modules for restoration vehicles
- Developed CAD models for plasma-cut components, ensuring precise fit and durability under performance-crucial conditions
- Implemented manufacturing processes installation procedures for various electrical, reducing manufacturing and installation time by 30%
- Managed electrical component procurement and inventory, optimizing for manufacture costs

PROJECTS

Graph of Experts Model

Atlanta, GA

Research

September 2025 – Ongoing

- Proposed modular mixture-of-experts architecture with expert traversal modeled as a trainable graph
- Implemented modality-agnostic encoder, graph router, expert bank, and decoder with TensorFlow
- Authoring working paper "Graph-of-Experts: Trainable Graphs for Hierarchical Multi-Scale Reasoning", formalizing model math and training objectives
- Working on applying GoE to Vesuvius scroll problem as a multi-scale case study (tokenization design, auxiliary routing feature design), geometry/ink segmentation workflow)
- Implemented fast parallelized data download and processing via thread pool executor

Blam

Atlanta, GA

Software Developer

May 2025 – Ongoing

- Experimental programming language combining Python-like ergonomics with Rust-style safety
- Authored formal EBNF grammar, tokenizer, and semantic analyzer, and implemented core language features (first-class functions, dynamic typing, arrays/vectors, ‘Bloop’ package manager)
- Implemented minimal IR code generator using LLVM
- Designed standard project layout and documentation
- Roadmap includes bootstrapper completion and richer code generation

Personal Website

Atlanta, GA

Web Developer

December 2024 – February 2025

- Developed modern responsive site via Next.js, Tailwind, and Framer for my personal portfolio and information
- Modeled 3D objects with Blender, using Three.js for responsive animations, and implemented Perlin background via WebGL

Djinn Code Generation CLI

Atlanta, GA

Software Developer

December 2024 – February 2025

- Rust-based CLI for automatic boilerplate generate with flexible API for natural language conversion via templating